Overview of materials for Polyamide-Imide, Glass Filled

Categories: Polymer; Thermoplastic; Polyamide-imide (PAI); Polyamide-Imide, Glass Filled

Material Notes: This property data is a summary of similar materials in the MatWeb database for the category "Polyamide-Imide, Glass Filled". Each property range of values reported is minimum and maximum values of appropriate MatWeb entries. The comments report the average value, and number of data points used to calculate the average. The values are not necessarily typical of any specific grade, especially less common values and those that can be most affected by additives or processing methods.

Physical Properties	Metric	English	Comments
Density	1.58 - 1.90 g/cc	0.0571 - 0.0686 lb/in³	Average value: 1.65 g/cc Grade Count:7
Water Absorption	0.200 - 0.520 %	0.200 - 0.520 %	Average value: 0.284 % Grade Count:4
Water Absorption at Saturation	1.30 - 1.50 %	1.30 - 1.50 %	Average value: 1.45 % Grade Count:4

Mechanical Properties	Metric	English	Comments
Hardness, Shore D	90.0 - 92.0	90.0 - 92.0	Average value: 90.7 Grade Count:3
Tensile Strength, Ultimate	115 - 221 MPa	16700 - 32100 psi	Average value: 154 MPa Grade Count:6
Elongation at Break	2.00 - 3.20 %	2.00 - 3.20 %	Average value: 2.55 % Grade Count:4
Modulus of Elasticity	6.21 - 14.5 GPa	900 - 2100 ksi	Average value: 10.2 GPa Grade Count:4
Flexural Yield Strength	145 - 333 MPa	21000 - 48300 psi	Average value: 214 MPa Grade Count:5
	181 - 181 MPa @Temperature 232 - 232 °C	26300 - 26300 psi @Temperature 450 - 450 °F	Average value: 181 MPa Grade Count:1
Flexural Modulus	5.66 - 13.8 GPa	821 - 2000 ksi	Average value: 9.32 GPa Grade Count:4
	9.86 - 9.86 GPa @Temperature 232 - 232 °C	1430 - 1430 ksi @Temperature 450 - 450 °F	Average value: 9.86 GPa Grade Count:1
Compressive Yield Strength	40.0 - 345 MPa	5800 - 50000 psi	Average value: 152 MPa Grade Count:5
Compressive Modulus	3.79 - 7.93 GPa	550 - 1150 ksi	Average value: 5.75 GPa Grade Count:3
Izod Impact, Notched	0.400 - 1.07 J/cm	0.750 - 2.00 ft-lb/in	Average value: 0.687 J/cm Grade Count:4

Electrical Properties	Metric	English	Comments
Surface Resistance	1.00e+13 - 1.00e+18 ohm	1.00e+13 - 1.00e+18 ohm	Average value: 2.00e+17 ohm Grade Count:5
Dielectric Constant	3.80 - 6.30	3.80 - 6.30	Average value: 4.45 Grade Count:5
Dielectric Strength	17.7 - 33.0 kV/mm	450 - 838 kV/in	Average value: 25.9 kV/mm Grade Count:6
Dissipation Factor	0.00500 - 0.0500	0.00500 - 0.0500	Average value: 0.0210 Grade Count:4

Thermal Properties	Metric	English	Comments
CTE, linear	12.6 - 46.8 µm/m-°C	7.00 - 26.0 µin/in-°F	Average value: 30.1 µm/m-°C Grade Count:7
	40.0 - 40.0 µm/m-°C @Temperature >=150 °C	22.2 - 22.2 µin/in-°F @Temperature >=302 °F	Average value: 40.0 µm/m-°C Grade Count:1
Thermal Conductivity	0.360 - 0.360 W/m-K	2.50 - 2.50 BTU-in/hr-ft^2-°F	Average value: 0.360 W/m-K Grade Count:4
Maximum Service Temperature, Air	250 - 270 °C	482 - 518 °F	Average value: 260 °C Grade Count:4
Deflection Temperature at 1.8 MPa (264 psi)	268 - 282 °C	515 - 540 °F	Average value: 276 °C Grade Count:6
Glass Transition Temp, Tg	275 - 280 °C	527 - 536 °F	Average value: 277 °C Grade Count:4
Flammability, UL94	V-0	V-0	Grade Count:3